

Control Group Selection Affects Biomarker Discovery in Colorectal Cancer

A Comparative Analysis of TCGA COAD Tumor Against NAT and GTEx Normal Colon Tissues, with Independent External Validation

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Abstract

Identifying robust biomarkers for colorectal cancer (CRC) from transcriptomic data requires a well-defined normal control group, yet the choice between tumor-adjacent normal tissue (NAT) and healthy tissue from independent cohorts such as GTEx remains underexplored. In this study, we systematically compared differentially expressed genes (DEGs) identified from TCGA COAD tumor samples ($n = 288$) against four distinct control groups: TCGA NAT ($n = 41$), GTEx colon all ($n = 308$), GTEx Colon Transverse ($n = 167$), and GTEx Colon Sigmoid ($n = 141$). Batch effects between TCGA and GTEx cohorts were corrected using *neuroCombat*. DEG analysis was performed using Welch's t-test with Benjamini–Hochberg correction ($|\log_2FC| > 1$, $FDR < 0.05$). Machine learning classifiers — LASSO logistic regression and Random Forest — were applied to each comparison, achieving perfect cross-validated AUC (1.000) across all four settings. To rule out feature-selection leakage, classification was re-evaluated under a nested cross-validation scheme in which DEG selection was performed exclusively within training folds; corrected AUC remained 1.000 ($\Delta = 0.000$), confirming that the perfect separation reflects genuine biological signal rather than methodological artifact. A four-step biomarker stability analysis identified six candidate genes present in three or more comparisons; after removing two artifacts driven by batch-effect residuals (ENSG00000279473) and pseudogene annotation (HMGN1P36), four stable, biologically validated biomarkers were confirmed: **CDH3**, **CLDN1**, **ETV4**, and **KRT80**. All four genes were independently validated in an external cohort (GSE156451, $n = 72$ paired CRC patients), showing significant tumor upregulation ($\log_2FC$ range +2.6 to +4.1, all $p < 10^{-30}$). These genes are enriched in epithelial differentiation pathways and each independently validated in prior CRC literature. Critically, Venn diagram analysis demonstrated that NAT-based DEGs overlap only partially with GTEx-based DEGs, indicating that control group choice substantially shapes the biomarker landscape. This study provides a methodological framework for evaluating biomarker stability across normal reference definitions and independently replicates a previously published three-gene CRC classification panel (**CDH3**, **ETV4**, **CLDN1**).

Keywords: colorectal cancer, TCGA COAD, GTEx, differential expression, biomarker stability, LASSO, Random Forest, external validation, **CDH3**, **CLDN1**, **ETV4**, **KRT80**

1. Introduction

Colorectal cancer (CRC) is the third most commonly diagnosed cancer and the second leading cause of cancer-related mortality worldwide, with approximately 1.9 million new cases and 935,000 deaths recorded in 2020 [1]. Early detection and precise molecular stratification remain critical unmet needs, as five-year survival rates drop from over 90% for localized disease to below 15% for distant metastatic CRC [2].

Transcriptomic profiling through RNA sequencing has become a cornerstone of cancer biomarker discovery. Public repositories such as The Cancer Genome Atlas (TCGA) and the Genotype-Tissue Expression (GTEx) project provide large-scale, multi-institutional gene expression datasets that enable powered statistical comparisons across thousands of samples. However, a fundamental methodological question persists: which “normal” tissue should serve as the reference control when identifying cancer-associated differential expression?

Two reference choices are commonly employed. The first is tumor-adjacent normal tissue (NAT), which consists of non-cancerous tissue collected from the same patient and sequencing run as the tumor. NAT is biologically proximal to the tumor microenvironment and shares the same technical processing pipeline, minimizing batch effects. However, mounting evidence suggests that NAT is not truly normal — it exhibits intermediate expression profiles between healthy and cancerous tissue, particularly in the colon, and may carry field cancerization effects [3, 4]. The second choice is healthy tissue from independent cohorts such as GTEx, which represents true physiological baseline expression but introduces cross-cohort batch effects from differences in tissue procurement, RNA extraction, and sequencing protocols.

Prior studies in glioma, breast, and lung cancers have demonstrated that using GTEx as a reference captures a broader set of cancer-associated DEGs compared to NAT, with less than 60% overlap between the two gene lists [3]. For colon adenocarcinoma (COAD), where NAT exhibits pronounced transcriptomic similarity to tumor relative to other cancer types, this discrepancy is expected to be especially pronounced.

In this study, we addressed the following research question: *Does the choice of normal control group — NAT versus different GTEx colon tissue subtypes — substantively alter the set of DEGs and downstream biomarker candidates identified for TCGA COAD?* We further assessed whether stable biomarkers consistent across multiple control definitions could be identified and biologically validated, whether such biomarkers are corroborated by independent prior literature, and whether they replicate in a fully independent external CRC cohort.

2. Methods

2.1 Data Acquisition

Gene expression data were obtained from a harmonized TCGA–GTEx dataset available through the UCSC Xena platform, containing RSEM-normalized $\log_2(\text{TPM} + 0.001)$ values for 60,498 genes across 19,131 samples. COAD-relevant samples were selected using curated sample ID files: TCGA COAD Tumor ($n = 288$, suffix -01), TCGA COAD NAT ($n = 41$, suffix -11), GTEx Colon Transverse ($n = 167$), and GTEx Colon Sigmoid ($n = 141$). Sample group labels were assigned based on TCGA barcode suffixes and GTEx tissue metadata.

2.2 Batch Effect Correction

Prior to cross-cohort comparisons, systematic technical differences between TCGA and GTEx were visualized using principal component analysis (PCA). Batch effects were corrected using neuroCombat [5], treating cohort origin (TCGA vs. GTEx) as the batch variable with no biological covariates specified. Correction was applied to the subset of NAT and GTEx samples ($n = 349$). PCA was repeated after correction to confirm successful harmonization (Figure QC-1, QC-2).

Tumor samples were deliberately excluded from the ComBat correction. Including the tumor group in the same correction as the normal samples risks attributing genuine tumor-versus-normal biological variation to the batch term, which would shrink the very signal the downstream analysis is designed to detect. By correcting only the NAT + GTEx normal samples to a common scale and then comparing the unmodified tumor samples against this harmonized normal baseline, the biological tumor signal is preserved. This design choice is revisited in the Limitations section, as it introduces an asymmetry between the corrected normal samples and the uncorrected tumor samples.

2.3 Differential Expression Analysis

Four pairwise DEG analyses were performed, summarized in Table 1.

No.	Tumor group	Control group	Expression data
1	TCGA COAD Tumor (n=288)	TCGA NAT (n=41)	Original TPM
2	TCGA COAD Tumor (n=288)	GTEx All (n=308)	ComBat-corrected
3	TCGA COAD Tumor (n=288)	GTEx Colon Transverse (n=167)	ComBat-corrected
4	TCGA COAD Tumor (n=288)	GTEx Colon Sigmoid (n=141)	ComBat-corrected

Table 1. The four tumor-versus-normal comparisons. The tumor group is held constant; only the normal reference changes.

For each comparison, Welch's two-sample t-test (unequal variances) was applied gene-wise. $\log_2$ fold change ($\log_2FC$) was computed as the difference of group means in $\log_2$ space. Multiple testing was corrected using the Benjamini–Hochberg (BH) procedure [20]. Genes with $|\log_2FC| > 1$ and FDR-adjusted p-value < 0.05 were designated as significant DEGs. Volcano plots were generated for each comparison.

2.4 Machine Learning Classification

To evaluate the discriminative capacity of DEG-derived gene sets, a machine learning pipeline was applied to each comparison:

- **Feature selection:** The top 500 DEGs ranked by adjusted p-value were selected as the feature pool.
- **LASSO logistic regression (classification mode):** L1-penalized logistic regression was fitted using LogisticRegressionCV with 30 candidate regularization values ($C \in [10^{-3}, 10^2]$), optimized by 5-fold stratified cross-validation for AUC.
- **LASSO logistic regression (biomarker discovery mode):** A fixed regularization ($C = 0.5$) was applied to identify a broader set of candidate biomarker genes, enabling biologically informative gene selection beyond the minimum classifier.
- **Random Forest:** Random Forest classifiers (300 trees, balanced class weight) were trained on the same feature pool. Feature importances (mean decrease in Gini impurity) were computed and the top 50 genes identified.
- **Consensus biomarkers:** Genes selected by LASSO ($C = 0.5$) AND in the RF top-50 were defined as per-comparison biomarker candidates.

2.5 Leakage-Controlled Validation (Nested Cross-Validation)

Because the feature pool in Section 2.4 was selected using the full dataset before the train/test split, the reported classification performance is potentially subject to feature-selection leakage: information from the held-out samples could influence which genes enter the model, inflating apparent accuracy. To test for this, the entire pipeline was re-run under a nested cross-validation scheme. In each outer fold, the train/test split was performed first, DEG selection (Welch's t-test + BH correction) was computed using only the training samples, and LASSO and Random Forest were then fitted on those training-fold-derived features and evaluated on the untouched test fold. Corrected AUC values were compared against the original (leaked) values to quantify the magnitude of any inflation.

2.6 Biomarker Stability Analysis

A four-step stability framework was applied across the four comparisons:

- **Venn diagram construction:** Four-set and pairwise Venn diagrams were generated to visualize overlap between per-comparison candidate sets.
- **Stability score calculation:** Each gene was assigned a score equal to the number of comparisons in which it appeared in the consensus biomarker set.
- **Stratification by score:** Genes were grouped by stability score (1–4) and tabulated with direction and $\log_2\text{FC}$ values.
- **Final candidate selection:** Genes with stability score ≥ 3 were retained and subjected to biological validation.

2.7 Biological Validation

Gene name mapping: Ensembl IDs were mapped to HGNC gene symbols using the mygene.info REST API.

Pathway enrichment: Gene symbols of final candidates were submitted to g:Profiler (version e111) using GO:BP, GO:MF, GO:CC, KEGG, REAC, and WP sources with FDR < 0.05 threshold.

Survival analysis: Overall survival (OS) data for TCGA COAD patients were downloaded from the NCI Genomic Data Commons (GDC) API. Kaplan–Meier curves were estimated using the lifelines library and group differences assessed with the log-rank test.

2.8 External Validation

To test whether the final biomarker candidates generalize beyond the TCGA discovery cohort, an independent RNA-seq dataset (GSE156451) was obtained from the NCBI Gene Expression Omnibus. This dataset comprises 72 paired colorectal cancer patients (tumor and matched adjacent normal tissue) from an independent institution, quantified as FPKM. Crucially, this cohort shares no samples, sequencing center, or processing pipeline with the TCGA–GTEx discovery data, providing a stringent external test. Expression of the four final biomarkers was compared between tumor and normal samples using both an unpaired Welch's t-test and a paired t-test (to account for within-patient correlation), with $\log_2\text{FC}$ computed in the same direction convention as the discovery analysis.

2.9 Software

All analyses were performed in Python 3.x using pandas, numpy, scipy, scikit-learn, neuroCombat, matplotlib, matplotlib-venn, GEOparse, and lifelines libraries. Scripts are available in the scripts/ directory of the project repository (github.com/lanlanq2144-collab/COAD_control_stability_project).

3. Results

3.1 Sample Composition and Batch Effect Correction

A total of 637 samples were included: 288 TCGA COAD tumor, 41 TCGA COAD NAT, 167 GTEx Colon Transverse, and 141 GTEx Colon Sigmoid. PCA prior to batch correction revealed a strong cohort-driven separation, with GTEx samples clustering distinctly from TCGA samples along PC1 (Figure QC-1). Following neuroCombat correction applied to the NAT + GTEx subset ($n = 349$), the inter-cohort separation was substantially reduced while within-group biological structure was preserved (Figure QC-2).

3.2 Differential Expression Analysis

Across all four comparisons, large numbers of significant DEGs were identified ($|\log_2FC| > 1$, $FDR < 0.05$), summarized in Table 2.

Comparison	Up-regulated	Down-regulated	Total DEGs
Tumor vs. NAT	5,454	6,169	11,623
Tumor vs. GTEx All	7,659	9,903	17,562
Tumor vs. GTEx Transverse	7,105	9,411	16,516
Tumor vs. GTEx Sigmoid	8,387	10,911	19,298

Table 2. Differential expression results across the four control-group comparisons.

The GTEx-based comparisons consistently produced more DEGs than the NAT comparison (~42–66% more total), consistent with NAT exhibiting an intermediate transcriptomic state between healthy and tumor tissue. Volcano plots for each comparison are shown in Figures DEG-1 through DEG-4 (results/deg/).

3.3 Machine Learning Classification Performance

LASSO logistic regression with cross-validated optimal C values selected 1–3 genes per comparison, all achieving cross-validated AUC = 1.000. Random Forest likewise achieved AUC = 1.000 across all four settings, confirming that the top DEG feature set provides near-perfect discriminative separation between tumor and normal tissue regardless of control group (Figure ML-1). Per-comparison details are given in Table 3.

Comparison	LASSO-opt	LASSO-bio	Consensus	LASSO AUC	RF AUC
Tumor vs. NAT	3	19	7	1.000	1.000
Tumor vs. GTEx All	1	40	12	1.000	1.000
Tumor vs. GTEx Transverse	1	32	14	1.000	1.000

Comparison	LASSO-opt	LASSO-bio	Consensus	LASSO AUC	RF AUC
Tumor vs. GTEx Sigmoid	1	34	14	1.000	1.000

Table 3. Per-comparison machine learning results. LASSO-opt = genes selected at cross-validated optimal regularization; LASSO-bio = genes selected in biomarker-discovery mode ($C = 0.5$); Consensus = LASSO-bio $\cap$ Random Forest top-50.

3.4 Leakage-Controlled Validation

A perfect AUC of 1.000 raises the legitimate concern that performance is inflated by feature-selection leakage. To address this directly, classification was repeated under nested cross-validation, with DEG selection confined to the training folds of each split (Section 2.5). The corrected AUC values are shown in Table 4.

Comparison	Original AUC	Nested-CV AUC	Δ
Tumor vs. NAT	1.000	1.000	0.000
Tumor vs. GTEx All	1.000	1.000	0.000
Tumor vs. GTEx Transverse	1.000	1.000	0.000
Tumor vs. GTEx Sigmoid	1.000	1.000	0.000

Table 4. Leakage check. Corrected AUC is identical to the original across all comparisons.

In every comparison, the corrected AUC remained 1.000, identical to the original ($\Delta = 0.000$). Even when as few as one to nine genes were selected independently within each training fold, those genes perfectly classified the held-out test fold. This demonstrates that the perfect separation is not an artifact of leakage but reflects the genuinely large transcriptomic distance between CRC tumor and normal colon tissue. By analogy, distinguishing individuals 200 cm tall from those 150 cm tall by height succeeds whether or not the threshold is chosen using the test data.

3.5 Biomarker Stability Across Control Groups

The four-set Venn diagram (Figure STAB-1) revealed limited overlap among per-comparison biomarker candidate sets. Of 30 unique candidate genes identified across all comparisons, only 6 appeared in 3 or more comparisons (stability score ≥ 3), and none appeared in all 4. Pairwise overlaps were markedly asymmetric: the NAT candidate set shared only 1–2 genes with each GTEx set, whereas the three GTEx sets shared 5–8 genes with one another. The distribution of stability scores is shown in Table 5.

Score	Genes (n)	Interpretation
4 / 4	0	Consistent across all control groups
3 / 4	6	Consistent across 3 control groups (retained)
2 / 4	5	Moderately stable
1 / 4	19	Control-group-specific

Table 5. Stability score distribution across 30 unique candidate genes.

The stability heatmap (Figure STAB-2) shows that biomarker composition is substantially different

between the NAT comparison and the three GTEx comparisons, confirming that normal reference choice is a major determinant of the resulting biomarker set. Of the 6 genes with stability score ≥ 3 , two were excluded on technical grounds (Section 3.7).

3.6 Final Biomarker Candidates

After artifact exclusion, four high-confidence biomarker candidates were retained (Figure STAB-3), summarized in Table 6.

Gene	Full name	Stability	log ₂ FC vs NAT	log ₂ FC vs GTEx
CDH3	Cadherin-3 (P-Cadherin)	3 / 4	+4.84	+6.05
CLDN1	Claudin-1	3 / 4	+4.84	+6.43
ETV4	ETS Variant Transcription Factor 4	3 / 4	+5.35	+6.43
KRT80	Keratin 80	3 / 4	+7.14	+6.91

Table 6. Final biomarker candidates. All four are consistently up-regulated in tumor relative to both NAT and GTEx normal colon tissue.

3.7 Artifact Identification and Exclusion

Critical examination of the two excluded candidates revealed a diagnostic pattern. Both genes exhibited enormous log₂FC values against GTEx (> 9.8) but negligible differences against NAT (< 0.4):

- **ENSG00000279473**: log₂FC vs. NAT = +0.37 (barely significant), log₂FC vs. GTEx = +11.6. Querying the Ensembl REST API returned “ID not found”, confirming this gene ID was retired in current Ensembl releases. Its pattern is consistent with a pipeline-level quantification artifact from an older annotation version.
- **HMG1P36 (ENSG00000235734)**: log₂FC vs. NAT = +0.31, log₂FC vs. GTEx = +9.8. This non-coding pseudogene on chromosome 2q11.2 has no known functional characterization. Its concordance with NAT (both TCGA-derived) but extreme divergence from GTEx strongly implicates incomplete ComBat correction.

This analysis demonstrates that inclusion of an NAT comparison serves as a critical internal sanity check for distinguishing biologically meaningful DEGs from cohort-specific technical artifacts.

3.8 Pathway Enrichment Analysis

Submitting the four validated gene symbols (CDH3, CLDN1, ETV4, KRT80) to g:Profiler returned 119 significant terms (FDR < 0.05) across GO, KEGG, and Reactome databases (Figure BIO-1). The most significantly enriched biological processes were:

- Skin and epidermis development (GO:BP, $p = 1.82 \times 10^{-5}$)
- Epithelial cell differentiation (GO:BP, $p = 1.70 \times 10^{-4}$)
- Keratinocyte differentiation (GO:BP, $p = 1.89 \times 10^{-4}$)

This functional profile is consistent with the known biology of the identified genes: **CLDN1** is a tight junction protein critical to epithelial barrier integrity [11], **CDH3** mediates cell–cell adhesion in epithelial layers [7], **KRT80** is a cytoskeletal keratin expressed in intestinal epithelium [14], and **ETV4** drives epithelial–mesenchymal transition (EMT) in CRC [10].

3.9 Survival Analysis

Kaplan–Meier analysis using GDC-downloaded OS data ($n = 258$ matched tumor samples) did not identify statistically significant survival differences for any of the four biomarker genes when patients were split by median expression (all log-rank $p > 0.05$, Figure BIO-2). This null result is not unexpected: a simple median-split Kaplan–Meier analysis is insensitive. Prior publications demonstrating survival associations for CLDN1 [11] and KRT80 [14] employed multivariate Cox proportional hazards regression with clinical covariates, which was beyond the scope of the present analysis.

3.10 External Validation in an Independent Cohort

The four final biomarkers were tested in GSE156451, an independent cohort of 72 paired CRC patients sharing no samples or processing pipeline with the discovery data (Section 2.8). All four genes were significantly and concordantly up-regulated in tumor tissue (Table 7).

Gene	log ₂ FC (tumor)	Unpaired p	Paired p
CDH3	+3.16	1.1×10^{-36}	7.6×10^{-34}
CLDN1	+3.35	7.8×10^{-35}	5.5×10^{-33}
ETV4	+4.08	2.8×10^{-44}	2.9×10^{-37}
KRT80	+2.61	1.5×10^{-31}	1.9×10^{-28}

Table 7. External validation in GSE156451 ($n = 72$ paired patients). All four biomarkers replicate with the same direction of effect as the discovery cohort.

The direction of effect was identical to the discovery analysis for all four genes, although the magnitudes were attenuated (discovery log₂FC +4.8 to +7.1 vs. NAT; external +2.6 to +4.1). Such attenuation is expected when moving to an independent platform and cohort. The paired test, which controls for within-patient variability, remained highly significant for every gene, providing strong evidence that the four-gene panel captures a reproducible, cohort-independent feature of colorectal tumor biology rather than a peculiarity of the TCGA–GTEx data. Boxplots are shown in Figure EXT-1 (results/external_validation/).

4. Discussion

4.1 Control Group Choice Substantially Alters the Biomarker Landscape

The primary finding of this study is that the choice of normal reference tissue meaningfully changes which genes are identified as differentially expressed in CRC. The NAT comparison identified 11,623 DEGs, while GTEx-based comparisons identified 16,516–19,298 DEGs — a 42–66% increase. Venn diagram analysis showed that a substantial proportion of candidates from the NAT comparison do not appear in any GTEx comparison, and vice versa. This is consistent with prior observations that TCGA NAT in colon cancer occupies an intermediate transcriptomic state due to field cancerization [3, 4], and supports the use of GTEx-derived healthy tissue as a more biologically representative reference.

4.2 A Stability-Based Filtering Framework Removes Control-Group-Specific Noise

The stability scoring approach introduced here provides an objective, data-driven means to identify biomarkers robust to the definition of “normal.” By requiring consistent selection across ≥ 3 of 4 control group definitions, we reduced the initial pool of 30 candidate genes to 6 (score ≥ 3), then to 4 after artifact exclusion. This approach is analogous in spirit to cross-dataset validation strategies [6], but applied within a single study by varying the control group rather than using independent patient cohorts. The subsequent external validation in GSE156451 (Section 4.4) then provides the complementary across-cohort confirmation.

4.3 Perfect Classification Reflects Biology, Not Leakage

The uniformly perfect AUC of 1.000 initially appears suspicious, and an obvious concern is feature-selection leakage — selecting DEGs on the full dataset before splitting. The nested cross-validation analysis (Section 3.4) directly refutes this explanation: when DEG selection was confined to training folds, AUC remained exactly 1.000 ($\Delta = 0.000$). The transcriptomic gulf between colorectal tumor and normal colon tissue is simply large enough that a handful of top genes separate the classes perfectly regardless of how features are chosen. This is consistent with the equally strong, independently derived external-cohort effect sizes and indicates that the perfect discrimination is a property of the underlying biology rather than of the analysis design.

4.4 Independent Replication of a Known CRC Biomarker Panel and External Cohort Confirmation

A particularly striking result is the independent replication of a previously published three-gene CRC classification panel. Guo et al. [6] applied machine learning to multiple public CRC datasets and identified **CDH3**, **ETV4**, and **CLDN1** as the top three genes capable of classifying CRC from normal tissue using only three features. Our analysis, conducted with a distinct methodology (DEG + LASSO + RF stability framework) on a different data composition (TCGA + GTEx), independently arrived at the same three genes as top biomarker candidates, with **KRT80** as an additional novel addition.

This convergence is reinforced by our own external validation: all four genes — including KRT80 — were significantly upregulated in the fully independent GSE156451 cohort, with paired-test significance surviving correction for within-patient variability. The combination of literature replication and direct external validation provides unusually strong evidence for the robustness of this panel. Each gene also has a well-established mechanistic rationale for involvement in CRC:

- **CDH3**: P-cadherin mediates homophilic cell adhesion and is aberrantly upregulated in CRC, promoting invasiveness and correlating with poor prognosis [7, 8]. CDH3 has been validated as a serum biomarker with AUC = 0.90 for detecting distant metastasis [8].
- **CLDN1**: Claudin-1 overexpression in CRC promotes cancer stem cell properties, EMT, chemoresistance to oxaliplatin, and interaction with the EPHA2 signaling axis [11, 12]. Meta-analyses confirm its prognostic value [11].
- **ETV4**: This ETS-family transcription factor is upregulated in colorectal adenocarcinoma relative to adenoma, correlates with lymphovascular invasion and lymph node metastasis, and is an independent predictor of shortened overall survival [10].
- **KRT80**: Keratin 80 promotes invasion and migration through interaction with PRKDC and the AKT signaling pathway [14]. siRNA-mediated knockdown suppresses CRC proliferation [15], and high KRT80 expression is an independent adverse prognostic factor.

4.5 Artifact Detection as a Methodological Contribution

The identification and exclusion of two false-positive candidates (ENSG00000279473 and HMGN1P36) highlights an underappreciated hazard in cross-cohort transcriptomic analyses. Both genes exhibited enormous $\log_2FC$ values against GTEx (> 9.8) but negligible differences against NAT (< 0.4). Since both NAT and tumor are TCGA-derived samples processed with identical pipelines, their concordance serves as an effective negative control. We recommend that future cross-cohort biomarker discovery studies routinely include a same-cohort normal reference as an internal control to identify and exclude such artifacts.

4.6 Limitations

Several limitations should be noted. First, ComBat correction was applied only to the normal samples and not to the tumor group (Section 2.2). While this preserves tumor biological signal, it creates an asymmetry in which the corrected GTEx normal samples and the uncorrected tumor samples are placed on a common scale only approximately; the two excluded artifact genes are direct evidence that this correction is imperfect at the individual-gene level. An alternative design that includes tumor in the correction with an explicit biological covariate would be a valuable robustness check. Second, the survival analysis used a simple median-split Kaplan–Meier test without multivariate adjustment for clinical covariates such as stage or microsatellite-instability status. Third, the small NAT cohort ($n = 41$) reduces statistical power and may inflate the apparent NAT-versus-GTEx divergence. Fourth, the machine learning analysis used a fixed top-500 DEG threshold, which was not itself tuned. Finally, all findings are observational and computational; the external validation strengthens generalizability but wet-laboratory and protein-level confirmation remain necessary before any clinical translation.

5. Conclusion

This study demonstrates that the choice of normal reference tissue — TCGA adjacent normal (NAT) versus GTEx healthy colon tissue — is a critical determinant of which genes are identified as differentially expressed in colorectal cancer and which biomarker candidates emerge from downstream machine learning analysis. GTEx-based comparisons consistently identified a larger and partially distinct set of DEGs compared to NAT, consistent with NAT representing a transcriptionally intermediate, field-cancerized tissue state. A stability-based framework requiring consistent biomarker selection across multiple control group definitions identified four high-confidence candidates — CDH3, CLDN1, ETV4, and KRT80 — all of which are upregulated in TCGA COAD tumor, are enriched in epithelial differentiation pathways, and have each been independently validated in prior CRC literature. Perfect classification performance was shown to be leakage-free via nested cross-validation, and all four biomarkers were independently confirmed in an external 72-patient cohort. Notably, the CDH3–CLDN1–ETV4 combination was independently replicated from a previously published machine learning-based CRC classification study. Inclusion of an NAT-based comparison as an internal control proved essential for identifying and excluding two batch-effect-driven false-positive candidates. These findings offer both a practical biomarker panel for CRC and a methodological framework for robust, control-group-agnostic biomarker discovery from public transcriptomic repositories.

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Data availability: All analysis scripts and result files are available in the project repository (github.com/lanlanq2144-collab/COAD_control_stability_project). External validation data: NCBI GEO accession GSE156451. Conflict of interest: None declared.